GRADE 10
MATHEMATICS
903
TIME $(84.8528137423857)^2$ Seconds

Name_____Grade_____

ADM No_____ Date_____ Sign_____

Instructions to Candidates
General Instructions

1. **You have 2hrs to complete this paper. Ensure you allocate your time wisely between the Multiple Choice and Structured sections.**
2. **Write your Name, Admission Number, and Class clearly in the spaces provided at the top of this page.**
3. **Electronic calculators may be used. Ensure your calculator is cleared of all stored data.**

Section: Structured and Essay Questions

1. **Answer all questions in the spaces provided below each question.**
2. **Show All Workings: Mathematical marks are awarded for correct methods and intermediate steps, even if the final answer is incorrect. Workings must be neat and legible.**
3. **Write your final answers in blue or black ink. Only diagrams and graphs should be drawn in pencil.**
4. **For calculations involving π use the value 3.142 or 22/7 unless a specific value is provided in the question.**
5. **Always state the correct units where applicable. Failure to include units may result in a loss of marks.**

Final Check

- **Cross out any rough work you do not wish to be marked with a single diagonal line.**
- **Ensure all extra sheets or graph papers are firmly attached to the main booklet.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. **Kawira formed the largest possible number using the digits 0, 7 and 9. She also formed the smallest number using the digits then found their sum. What was the answer? (4mks)**

2. **What is the least length possible which can be cut exactly into equal portions of 56cm, 63cm and 84cm long? (2mks)**

3. **Express $1.41\dot{6}$ as a mixed number (2mks)**

4. **The sum of two numbers is 20. If the difference in their squares is 120, find the two numbers (3mks)**

5. **The sum of the interior angles of a regular polygon is 2880°. Find the size of each exterior angle (3mks)**

6. **Find the distance between the minute hand and hour hand when the clock is at 2.25 pm (3mks)**

7. **Solve and express the following pair of simultaneous inequality and represent the solution as a combined inequality and express in a number line (4mks)**

$$2x + 5 > 5x - 4$$

$$4x - 3 \geq 6 - 5x$$

8. **Calculate the area of an equilateral triangle of side 25cm (3mks)**

9. **Common salt has a density of 2.2 g/cm^3 while sand has a density of 3.2g/cm^3. If 0.8 kg of salt is mixed with 1.5kg of sand, find the density of the mixture (4mks)**

10. **The longer parallel side of a trapezium is twice as long as the shorter parallel side. The perpendicular distance between**

the parallel sides is 10cm. If the area of the trapezium is 225cm^2, find the length of the longer parallel side (3cm)

11. Simplify (2mks)

$\bar{1}.57 - \bar{2}.11$

12. Solve (3mks)

$10^m = 3.6$

13. Find the quadratic equation whose roots are $x = -\frac{5}{2}$ $and\ x = 3$ (3mks)

14. Solve the equation (4mks)

$$log\ (2x + 1) + log\ (3 - x) - log\ 5\ = 0$$

15. Factorize completely (3mks)

$$36x^2 - 4y^2 + 7xy$$

16. Use tables only to work out

a) $\frac{5}{0.3786} + \frac{4}{0.462}$ **(3mks)**

b) $\left(\frac{5.86 \times 0.2432}{log\ 8}\right)^{\frac{3}{4}}$ **(3mks)**

c) $\sqrt[3]{\left(\frac{6.09}{5.67\times9.035}\right)}$ **(3mks)**

17. The line y=mx+5 makes an angle of 32˚ with the x-axis. If m is negative, find the coordinates of the x-intercept of the line. (3mks)

18. Boxes of mathematical sets that measure 16cm by 8cm and 2cm are packed in a carton. Each box is similar to the carton. If the linear scale factor between a box with a set and the carton is 3, find the volume of the carton (3mks)

19. The G.C.D of three numbers is 60 and their L.C.M is 6600. Two of the numbers are 120 and 600. Determine the least possible value that the third number can take. (3mks)

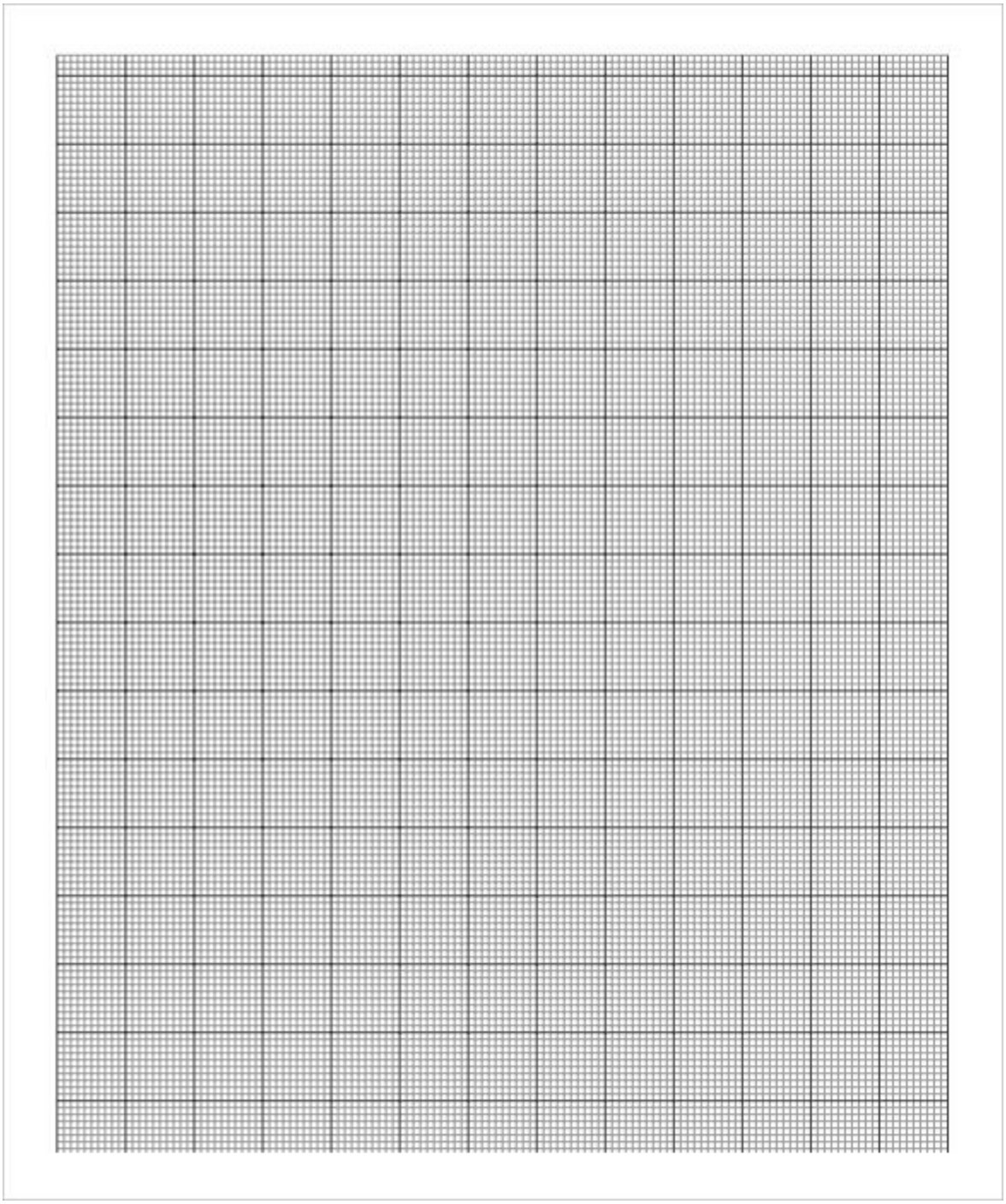
20. Cheruto has 55 notes whose total value is Sh. 11300. There are thrice as many Sh.200 notes as there are Sh.50 notes. If the rest are Sh.500 notes, how many more Sh.50 notes than Sh.500 notes does she have? (4mks)

21. Three business partners, Nafula, Nekesa and Nanjala are to share a profit Sh. 2014220 in the ratio 2:4:x respectively. If Nekesa received Sh.309880, determine the value of x. (3mks)

22. Express the following inequalities on a grid (4mks)

$$2x - 5y < 8$$

$$3y + 6y > 2$$



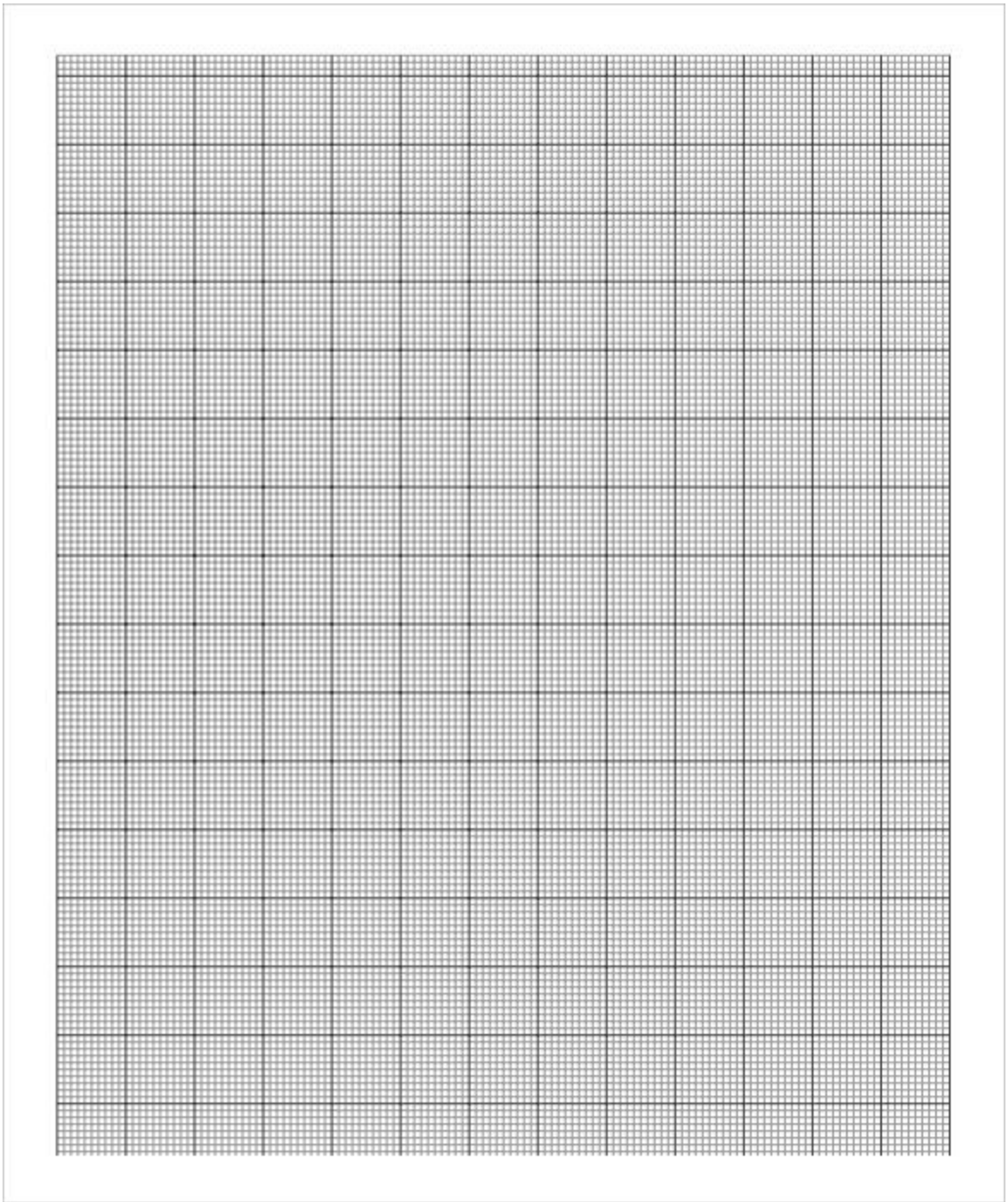
23. Use a ruler and a pair of compasses only to construct triangle ABC such that AB=2.5cm, BC= 3.5cm and AC= 5.5cm. Measure <ABC. (3mks)

b. Construct a circle whose circumference touches all the vertices and measure its radius (4mks)

c. Construct a perpendicular from A to a point T on CB produces (1mk)

d. With BT as the base, calculate the area of ABT (2mks)

24. The image of point P (-7,7) after an enlargement scale factor -3 is P` (5, -1). Find the coordinates of the center of the enlargement (4mks)



25. The masses to the nearest kg of 60 people in a blood donation Centre were recorded as shown in the table below

MASS (KG)	40-44	45-49	50-54	55-59	60-64	65-69	70-74	75-79	80-84
FREQUENCY	1	3	5	12	16	10	7	4	2

a. **Draw a new table showing the midpoint, cumulative frequency and fx. (1mk)**

b. **Find the mean class and mean (4mk)**

c. **Find the median class and hence calculate the median (5mk)**

d. **State the modal class(1mk)**